

Deriving, Testing, and Bounding the Multi-Power Law for Schedule-Aware Loss-Curve Prediction

Jiaju Wu (2400010711, Undergraduate)

Deep Learning — Task 2 | github.com/wjjpku/DL-final

One-line summary: we reproduce the two empirical loss-curve laws (Tissue, MPL), *derive* MPL from SGD dynamics (a discrete cousin of the kernel-regression functional scaling law of Li et al. 2025), confirm its predicted scale-invariance (all 27 curves collapse onto one master curve, $R^2=0.997$), try *four* principled ways to *beat* it (honest result: MPL is near-optimal), and give its one empirical parameter γ a corrected *edge-of-stability* origin (river-valley sharpening lag) while showing it is near-irreducible.

*These slides are written as a **self-contained report**: each slide states its setup, its result, and a boxed/bold **takeaway**, so they can be read without an oral presentation. Every number is reproduced by a named script (final two slides).*

Outline

- ① Problem & Motivation
- ② Background & Reproduction
- ③ Deriving MPL from SGD Dynamics
- ④ Scale Invariance & a Universal Master Curve
- ⑤ Method Development: Can We Improve MPL?
- ⑥ The Edge-of-Stability Origin of γ
- ⑦ Conclusion

The problem: schedule-aware loss-curve prediction

Goal. Given a learning-rate (LR) schedule $\{\eta_t\}$, predict the whole pretraining loss curve $L(t)$ *before* running the training.

Why it matters.

- Pretraining LLMs is expensive; the LR schedule strongly shapes the entire loss trajectory, not just the endpoint.
- Classical scaling laws (Kaplan, Chinchilla) predict only the *final* loss vs. model size / data — they say nothing about the curve's response to the schedule *shape* (warmup, decay form, horizon).
- A good law (i) replaces trial runs with a formula, (ii) exposes mid-training dynamics (the extra drop from LR decay), and (iii) can be inverted to *search* for better schedules.

The defining test — cross-schedule generalization: **fit on cosine schedules, predict the unseen WSD family.**

What this project does

We go beyond reproduction and study the governing law itself:

1. **Reproduce** the two representative laws (Tissue 2024; Multi-Power Law / MPL, Luo 2025) on the cosine→WSD task.
2. **Derive** MPL's functional form from SGD on a power-law curvature spectrum — giving each parameter a physical meaning and a prediction.
3. **Test** the prediction: kernel exponents are scale-invariant $\Rightarrow$ all curves collapse onto one master curve.
4. **Improve?** Four principled methods (SC-MPL; a new (S, Q) asymptotic; an explicit river-valley law; a sharpness-lag correction) — honest finding: MPL is near-optimal here.
5. **Bound:** give MPL's one empirical parameter γ a corrected *edge-of-stability* origin and show it is near-irreducible.

Data: official public loss curves, 3 scales (25/100/400M) $\times$ 9 schedules.

Background: the two empirical laws

Let $S(t) = \sum_{i \leq t} \eta_i$ be the **cumulative LR** (a natural “time”), and $\Delta\eta_k = \eta_{k-1} - \eta_k$ the LR decrement at step k .

Tissue et al. (5 parameters) — simple, interpretable, but coarse:

$$L(t) = L_0 + A S(t)^{-\alpha} - C S_2(t), \quad S_2 = \text{momentum-smoothed annealing area.}$$

Multi-Power Law / MPL (7 parameters) — current state of the art:

$$L(t) = L_0 + \underbrace{A S(t)^{-\alpha}}_{\text{backbone}} + \underbrace{B \sum_{k \leq t} \Delta\eta_k G(\eta_k^{-\gamma}(S(t) - S(k)))}_{\text{loss-drop term } LD}, \quad G(x) = 1 - (1 + Cx)^{-\beta}.$$

How to read MPL. The *backbone* $A S^{-\alpha}$ is the loss you would get at the current cumulative LR with no decay. The *loss-drop* term LD adds the bonus from every past LR decrease $\Delta\eta_k$: G is a **saturating** kernel (rises $0 \rightarrow 1$), so each decrement’s benefit accrues over “time” $S(t) - S(k)$ and then plateaus. The factor $\eta_k^{-\gamma}$ rescales that time by the LR at which the decrement happened — it is **fit empirically**, with no derivation in the original paper. *Pinning down this γ is the through-line of this project.*

Reproduction: setup

Setup (following the task).

- Data: official MPL loss curves, 25/100/400M, 9 schedules each.
- **Fit on cosine** schedules (cosine_24000, cosine_72000); **evaluate on 5 unseen WSD curves** (wsd, wsdld, wsdcon_{3,9,18}).
- Objective: Huber loss on $\log L$; optimizer: L-BFGS-B with multi-start.
- Engineering: Tissue's S_2 rewritten as an exact linear recurrence ($\sim 100\times$ faster); MPL's LD vectorized.

Metric. Average test MAE over the 5 WSD curves (lower is better).

Code: repro/reproduce_cosine_to_wsd.py.

Reproduction: results match the published numbers

Scale	Tissue (5p)	MPL (7p)
25M	0.00783	0.00539
100M	0.00486	0.00496
400M	0.00979	0.00852

avg test MAE on WSD (fit on cosine)

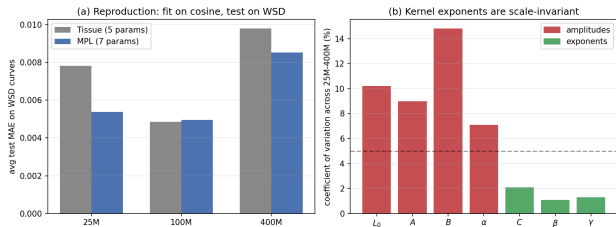


Fig. 1(a): predicted curves (lines) vs. ground-truth loss (dots) on held-out WSD schedules; closer is better.

Analysis (what the numbers say). We recover the published behavior: MPL is more accurate at 25/400M and ties Tissue at 100M, and generalizes more stably across schedule shapes. Our MPL test MAE (0.0054/0.0050/0.0085 at 25/100/400M) matches Luo et al., confirming the reproduction is faithful. **This well-fit MPL is the baseline the rest of the work must respect** — every later “improvement” is measured against a properly-optimised MPL, not a weak one.

Why derive it? From 7 opaque parameters to physical meaning

Goal of this section. MPL fits well but its 7 parameters are a black box. We derive its *form* from a simple dynamical model so each parameter gets a physical identity *and* a testable prediction about how it scales with model size.

Model. Near the trajectory the loss is locally quadratic. In the Hessian eigenbasis, mode i has curvature λ_i and gradient-noise variance σ_i^2 ; $\delta_i(t)$ is the distance to the optimum along that mode, and one SGD step is the linear noisy recurrence

$$\delta_i(t+1) = (1 - \eta_t \lambda_i) \delta_i(t) - \eta_t \xi_i(t), \quad L = L_{\min} + \frac{1}{2} \sum_i \lambda_i \mathbb{E}[\delta_i^2].$$

Each step shrinks mode i by $(1 - \eta_t \lambda_i)$ and injects noise. The mean square $\mathbb{E}[\delta_i^2]$ splits into a **bias** (deterministic shrinkage of the initial gap) and a stochastic **noise floor**.

(1) Bias $\rightarrow$ power-law backbone. The bias shrinks as $\delta^{\text{bias}} \approx \delta(0) e^{-\lambda S}$, governed by the **cumulative LR** $S(t) = \sum_{i \leq t} \eta_i$ (high-curvature modes die first; flat modes persist). Summing over a power-law density of curvatures $g(\lambda) \sim \lambda^a$, a Tauberian (Laplace-transform) argument turns the mixture of exponentials into a single power law:

$$L_{\text{bias}}(S) = \int_0^\infty g(\lambda) e^{-2\lambda S} d\lambda \sim A S^{-\alpha}, \quad \alpha = a + 1, \quad L_0 = L_{\min}.$$

Takeaway: the $A S^{-\alpha}$ backbone is just “many modes relaxing at a power-law spread of rates.”

The annealing term and a falsifiable prediction

(2) Noise floor $\rightarrow$ annealing term. The stochastic part obeys $V_i(t) \approx \sigma_i^2 \int_0^{S(t)} \eta(S') e^{-2\lambda_i(S(t)-S')} dS'$: the iterate carries a cloud of noise whose size tracks the *current* LR, so at constant LR the noise loss is $\propto \eta$ (**the floor tracks the current LR**). When the LR *decays*, this floor shrinks — that extra drop, written as a sum over the per-step LR *decrements* $\Delta\eta_k$, is

$$\Delta N(t) = \sum_k \Delta\eta_k \mathcal{K}(S(t) - S(k)) \quad \text{— this is exactly MPL's loss-drop term } LD.$$

Assuming a **Gamma-shaped** curvature spectrum $\propto \lambda^{\beta-1} e^{-\lambda/\lambda_0}$ makes the kernel $\mathcal{K}(y) \propto 1 - (1 + Cy)^{-\beta}$ with $C = 2\lambda_0$ — *precisely* MPL's G . So the whole form of MPL (backbone + LD) drops out of the dynamics.

Each parameter now has an identity:

Exponents $\{\alpha, \beta, C\}$ = functions of the spectral *shape* (the density's exponents)
Amplitudes $\{L_0, A, B\}$ = set by the irreducible loss + spectral *scale*

Falsifiable prediction. If the curvature spectrum keeps its *shape* as model size N grows (only its scale changes), then the **exponents must be scale-invariant** and only the amplitudes drift with N . We test this next.

Where this sits: theory, empirics, and learned laws

Our SGD–spectrum derivation is a **discrete-spectrum cousin** of the kernel-regression theory of **Li, Chen, Huang, Wang & Wu (2025)**, *Functional Scaling Laws* (FSL, NeurIPS’25):

	FSL (Li et al. 2025)	this work
model	SGD on power-law kernel regression	SGD on power-law Hessian spectrum
“time”	intrinsic time	cumulative LR $S(t) = \sum \eta_i$
schedule enters via	a convolutional functional	the decrement kernel LD
instantiated on	const / exp-decay / WSD (theory)	real MPL curves, 25/100/400M

The landscape of laws. *Tissue* (momentum law) and *MPL* (empirical) are closed forms; *FSL* is the matching kernel-regression theory; *NCPL* (Zhang, Wen & Ma 2026) instead **learns** a neural configuration $\rightarrow$ performance map (evaluated on the StepLaw dataset). **We take the interpretable closed-form route:** derive MPL, confirm its universality, and pin down its one irreducible parameter.

Result 1: the kernel exponents are scale-invariant

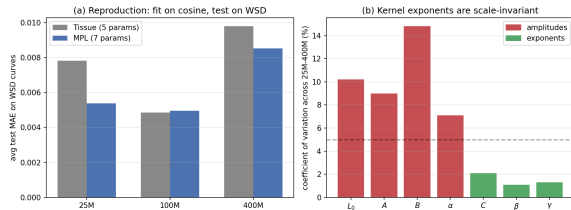


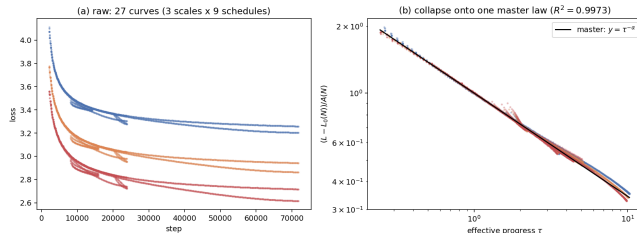
Fig. 1(b): each fitted parameter vs. model size. The exponent curves are flat; the amplitude curves are sloped.

Fitting MPL separately at each scale and comparing parameters across a $16\times$ size range (25 $\rightarrow$ 400M):

- Exponents C, β, γ : CV = 2.1%, 1.1%, 1.3% (essentially constant).
- Amplitudes L_0, A, B : CV = 9–15% (drift with N).
- α (initialization-weighted, predicted least stable of the exponents): 7%.

The prediction is confirmed: the shape of the annealing response is a near scale-invariant constant; only the amplitudes carry the size dependence.

Result 2: all 27 curves collapse onto one master curve



Top: 27 raw loss curves (3 sizes $\times$ 9 schedules). Bottom: after the τ -rescaling they fall on one line $y = \tau^{-\alpha}$.

Analysis. The collapse succeeds *only because a single exponent set fits every scale* — a direct visual proof of scale invariance. There is essentially **one** loss curve; model size and schedule merely reparameterize the axes.

Rearranging MPL algebraically shows it is really a *single* power law $L = L_0(N) + A(N) \tau^{-\alpha}$ in one **effective-progress** variable

$$\tau = [S^{-\alpha} + (B/A)LD]^{-1/\alpha},$$

which absorbs the entire schedule (backbone + annealing) into a single number per step — “how far along the universal curve are we.”

Using **one shared exponent set** for all scales and only per-scale amplitudes, all **27 curves** collapse onto $y = \tau^{-\alpha}$ with $R^2 = 0.997$.

Code: repro/universal_collapse.py

Attempt 1 — SC-MPL: a scale-conditioned law

Idea. Since the exponents are scale-invariant, **borrow them from cheap small models** and fit only the 3 amplitudes $\{L_0, A, B\}$ on the target scale — cutting a new scale from 7 free parameters to 3.

Scale	MPL	SC-4	SC-3
100M	0.00435	0.00451	0.00525
400M	0.00484	0.00760	0.00786

fair comparison, official split, avg test MAE

Honest result. Competitive at 100M (wins 4/6 curves with far fewer parameters) but **does not beat a well-fit MPL at 400M**: the residual drift (chiefly in α) costs more than the parameter economy gains.

A methodological caution we discovered. An earlier protocol that *re-fit* the MPL baseline into a poor optimum made SC-MPL look 20% better; giving the baseline its **true** optimum erases that “win.” *An under-optimized baseline can fabricate a false result.*

Attempt 2 — Q-MPL: a new discrete-step asymptotic

New angle. Tissue and MPL both use the continuum approximation, leaving the single time variable $S = \sum \eta$. Keeping the **next discrete-step order** introduces a *second* time variable:

$$\prod (1 - \eta\lambda)^2 \approx e^{-2\lambda S - \lambda^2 Q}, \quad Q = \sum \eta^2.$$

Expanding the bias integral yields a **derived correction** $\propto Q S^{-(\alpha+2)}$. We test Q-MPL = MPL + $E \cdot Q S^{-(\alpha+2)}$ (8 params).

Scale	MPL	Q-MPL	E^*
25M	0.00715	0.00715	≈ 0
100M	0.00386	0.00386	≈ 0
400M	0.00669	0.00824	overfit

Honest result. The data drive $E \rightarrow 0$ at 25/100M (the term is unnecessary), and when free it **over-fits cosine and worsens WSD** at 400M. **Q-MPL does not beat MPL.**

Verdict: MPL is near-optimal on the current data

Two *principled* extensions — scale-conditioning and a new (S, Q) asymptotic — both fail to improve a well-fit MPL (two more, built on the edge-of-stability mechanism, follow in §6 and also fail). The universal collapse ($R^2 \approx 0.997$) shows MPL already captures essentially all reproducible structure.

Honest conclusion

On these curves MPL is near-optimal: the residual ($\text{MAE} \sim 0.003$) carries no structure that a derived correction exploits.

Why, and what would change it. MPL was developed and tuned on exactly this kind of (relatively mild) data. Beating it would likely require data that **exposes MPL's failure modes** — more extreme schedules or more model scales — which is the natural data bottleneck of this study.

MPL's one empirical parameter: γ is essential, but not from theory

The factor $\eta_k^{-\gamma}$ is the only part of MPL our derivation does *not* produce. Is it actually needed?

Scale	full MPL	$\gamma=0$
25M	0.0032	0.0119
100M	0.0038	0.0144
400M	0.0049	0.0162

real data, test MAE

Setting $\gamma=0$ (re-fitting amplitudes) makes the prediction **3–4× worse** — so γ is **essential**.

Yet it *cannot* come from the spectral theory: in S -units the leading kernel has no η factor, so the model predicts $\gamma=0$, and a quadratic-SGD **simulation confirms** a $\gamma=0$ kernel fits the simulated curves as well as full MPL. So γ lives *outside* the quadratic picture. Where?

The corrected origin: an *adaptive* river valley

Fixing a common misconception. The data are trained with **AdamW**, not gradient descent — so the operative edge is *not* sharpness $= 2/\eta$, but the **adaptive edge of stability** (Cohen et al. 2022). AdamW pretraining traverses a **river-valley** landscape (Wen et al. 2024): a flat “river” (progress) and sharp “mountain” directions (oscillation).

For a mountain mode of curvature H , the stationary excess loss of discrete SGD is *exactly*

$$P_{\text{osc}}(\eta, H) = \frac{\sigma^2}{2} \frac{\eta}{2 - \eta H}.$$

- $\eta H \rightarrow 2$ (the edge) $\Rightarrow P_{\text{osc}} \rightarrow \infty$: the **elevated stable-phase loss**.
- $\eta \rightarrow 0$ (decay) $\Rightarrow P_{\text{osc}} \rightarrow 0$: the decay **“reveals” accumulated progress**.
- H is a **lagging state** (progressive sharpening, timescale τ). If H tracked η instantly, $P_{\text{osc}} \propto \eta$ and *no* γ is needed; a finite lag after a **sharp** decay makes the penalty history-dependent — exactly what $\eta_k^{-\gamma}$ encodes.

γ = the effective exponent of the sharpening lag.

Two explicit laws — and why γ is near-irreducible

We replaced γ by the mechanism itself (theory first, then experiment):

(a) **RV-EoS** (explicit, 7 params, all physical):

$$L_0 + AS^{-\alpha} + \frac{B}{2} \frac{\eta_t}{2 - \eta_t H_t}, \quad H_{t+1} = H_t + \frac{1}{\tau} (u^* / \eta_t - H_t).$$

(b) **Sharpness-lag correction** on *frozen* MPL:

+ $\kappa(\tilde{\eta}_t - \eta_t)_+$, $\tilde{\eta}$ = EMA of η — **dormant on cosine** by construction.

law	MPL	new	wins
RV-EoS	0.0041	0.0097	0/15
lag-corr.	0.0036	0.0037	7/12

mean MAE, held-out WSD

RV-EoS is $2.4\times$ worse (u^* rails to the edge, τ not invariant). The correction **improves true end-of-training decays by 20–52%** — the residual is real — but over-corrects others and is not scale-invariant.

Result: γ is near-irreducible

The residual is a genuine edge-of-stability *sharpening lag*, but capturing it needs a schedule-specific sharpness *state* that cumulative LR cannot recover. Across **four** attempts, MPL's single γ is its most transferable compression.

Conclusion & takeaways

What we did.

- Reproduced Tissue & MPL (cosine→WSD), matching published accuracy.
- Derived MPL from an SGD–spectrum model, giving its parameters physical meaning and a falsifiable prediction.

Main findings.

- **Scale invariance & universality:** the kernel exponents are constant to 1–2%; all 27 curves collapse onto one master power law ($R^2=0.997$) — there is essentially one loss curve.
- **Honest method development:** *four* principled improvements (SC-MPL, Q-MPL, RV-EoS, sharpness-lag) do not beat a well-fit MPL; it is near-optimal on current data. We also show how an unfair baseline fabricates a false win.
- **The limit of the theory:** γ is essential yet underivable from quadratic theory; corrected for AdamW it is a *river-valley sharpening lag* — physically real but near-irreducible from cumulative-LR data.

Next steps. Resolve the sharpness *state* γ compresses (measured curvature traces, more extreme schedules & scales); connect to the FSL theory of Li et al. (2025).

Reproducibility

All results are reproduced by scripts in the repository:

Result	Command
Reproduction (Tab. 1, Fig. 1a)	<code>repro/reproduce_cosine_to_wsd.py</code>
Universal collapse (Fig. 2)	<code>repro/universal_collapse.py</code>
SC-MPL fair comparison	<code>repro/validate_theory.py</code>
Q-MPL new asymptotic	<code>repro/qmpl.py</code>
RV-EoS explicit river-valley law	<code>repro/river_valley.py</code>
Sharpness-lag correction	<code>repro/river_floor_lag.py</code>
γ ablation & simulation	<code>repro/validate_theory.py</code> , <code>sgd_spectrum_sim.py</code>
Derivations	<code>docs/core/derivation.md</code> , <code>river_valley_derivation.md</code>

```
pip install -r requirements.txt; data: the official MPL public loss curves (the task dataset), shipped in  
external/MultiPowerLaw/.
```

Division of labor & acknowledgments

Team. Single-member group: **Jiaju Wu** (2400010711, Undergraduate) — all work (reproduction, derivation, experiments, writing) by the sole author.

Resources used.

- Data: official Multi-Power-Law public loss curves (Llama-2-style 25/100/400M $\times$ 9 schedules).
- Key references: Tissue et al. (2024); Luo et al. (2024/25, MPL); Li, Chen, Huang, Wang & Wu (2025, FSL); Cohen et al. (2021/22, edge of stability); Wen et al. (2024, river valley); Zhang, Wen & Ma (2026, NCPL).

Code & docs: github.com/wjjpku/DL-final

Thank you! Questions welcome.